

IMMUNOLOGY

Original MCQ Reviewer

Concept-aligned questions based on Chapter 3 coverage

Includes answer key and rationale for every item

The questions and explanations in this reviewer are newly written. They test the same learning objectives and terminology as the supplied chapter without reproducing its question wording.

3.1 Basic Principles of Immunology

1. After a previously unexposed patient encounters a specific antigen, which substance represents a targeted adaptive immune product?

- A. Complement protein
- B. Lysozyme
- C. Immunoglobulin
- D. Normal commensal flora

2. Which pair serves as the primary lymphoid organs responsible for lymphocyte development and maturation?

- A. Spleen and lymph nodes
- B. Bone marrow and thymus
- C. Peyer patches and spleen
- D. MALT and thoracic duct

3. A mature B lymphocyte binds its specific antigen and becomes activated. Which cells can arise from this activated clone?

- A. Plasma cells and memory B cells
- B. Only neutrophils
- C. Cytotoxic T cells and NK cells
- D. Macrophages and dendritic cells

4. Which surface marker most specifically identifies a mature T lymphocyte?

- A. CD3
- B. CD19
- C. CD56
- D. CD14

5. A CD8-positive T cell recognizes a virally infected cell. The foreign antigen is normally presented with which molecule?

- A. MHC class I
- B. MHC class II
- C. IgM
- D. C3

6. Which complement component participates in both the classical and alternative pathways?

- A. C1
- B. C4
- C. C3
- D. Factor D

7. Which antibody class is usually produced first during a primary immune response?

- A. IgA
- B. IgE
- C. IgG
- D. IgM

8. Which immunoglobulin class normally provides passive humoral protection to the fetus by crossing the placenta?

- A. IgG
- B. IgM
- C. IgA
- D. IgE

3.2 Immunologic Procedures

9. The strength of binding between one antibody combining site and one antigenic epitope is called:

- A. Avidity
- B. Affinity
- C. Valency
- D. Specificity

10. A precipitation assay is unexpectedly negative in a specimen containing an extremely high antibody concentration. Which phenomenon should be suspected?

- A. Postzone
- B. Prozone
- C. Hemolysis
- D. Cross-reactivity

11. During an indirect ELISA, the technologist fails to remove unbound enzyme-labeled antibody adequately. What is the likely effect?

- A. Falsely low signal
- B. Falsely high signal
- C. No change
- D. Loss of antigen specificity only

12. An ELISA specimen produces absorbance above the highest calibrator. What is the most appropriate next step?

- A. Estimate beyond the curve
- B. Report the highest standard
- C. Dilute the specimen and repeat
- D. Use less substrate

13. Serum protein electrophoresis shows a narrow suspicious band. Which test is most useful to identify its heavy- and light-chain type?

- A. Immunofixation electrophoresis
- B. CBC
- C. Direct antiglobulin test
- D. Complement fixation

14. In a complement fixation assay, what does visible hemolysis in the indicator system signify?

- A. Positive test
- B. Negative test
- C. Prozone
- D. Antibody excess

15. What is the main purpose of gating during flow cytometry?

- A. Select the cell population to analyze
- B. Increase antibody affinity
- C. Lyse mature cells
- D. Remove fluorescence

3.3 Infectious Diseases

16. A reactive RPR is followed by a nonreactive treponemal-specific test. Which interpretation best fits this pattern?

- A. Confirmed active syphilis
- B. Biological false-positive nontreponemal test
- C. Tertiary syphilis
- D. Successfully treated syphilis

17. Fourth-generation HIV screening assays shorten the diagnostic window because they detect antibodies plus which viral marker?

- A. Reverse transcriptase
- B. p24 antigen
- C. Proviral DNA only
- D. gp120 antigen only

18. An HIV screening assay is repeatedly reactive, the HIV-1/2 differentiation assay is negative, and qualitative HIV RNA is positive. What does this pattern most strongly suggest?

- A. Early HIV infection before detectable antibody
- B. No HIV infection
- C. Past resolved infection
- D. Laboratory contamination is proven

19. Which laboratory method is preferred for diagnosing HIV infection in an infant born to an HIV-positive mother?

- A. Maternal antibody titer
- B. HIV nucleic acid testing
- C. Rapid IgG antibody test
- D. CD4:CD8 ratio alone

20. Which hepatitis B marker usually appears earliest during acute infection?

- A. Anti-HBs
- B. HBsAg
- C. Anti-HBe
- D. IgG anti-HBc

21. During the HBV core-window period, which antibody is especially useful for demonstrating recent infection?

- A. IgM anti-HBc
- B. Anti-HBs
- C. Anti-HBe only
- D. IgG anti-HBs

22. A patient has both IgM and IgG antibodies to EBV viral capsid antigen. Which interpretation is most consistent?

- A. Current or recent EBV infection
- B. Remote infection only
- C. No prior exposure
- D. Immunization response

23. When monitoring response to antiretroviral therapy, which measurement changes most directly with viral replication?

- A. HIV viral load
- B. RPR titer
- C. Total IgG
- D. Absolute neutrophil count

3.4 Autoimmune Diseases

24. Which concept best defines autoimmunity?

- A. Failure to recognize foreign antigen
- B. Loss of tolerance to self-antigens
- C. Complete absence of B cells
- D. Excessive complement synthesis

25. A high-titer anti-Smith antibody most strongly supports which diagnosis?

- A. Systemic lupus erythematosus
- B. Rheumatoid arthritis
- C. Systemic sclerosis
- D. Autoimmune hepatitis

26. Which ANA fluorescence pattern most often suggests antibodies to extractable nuclear antigens such as Sm, RNP, SSA, or SSB?

- A. Speckled
- B. Centromere only
- C. Peripheral only
- D. No nuclear staining

27. Rheumatoid factor most commonly consists of IgM directed against which portion of IgG?

- A. Fc region
- B. Fab region
- C. Light-chain variable region
- D. Hinge peptide only

28. A patient has inflammatory joint disease but repeated rheumatoid factor tests are negative. Which additional antibody test may support rheumatoid arthritis?

- A. Anti-CCP
- B. Anti-HBs
- C. Anti-mitochondrial antibody
- D. Anti-IgA

29. Which autoantibody is associated with both Graves disease and Hashimoto thyroiditis?

- A. Anti-thyroid peroxidase
- B. Anti-Sm
- C. Anti-CCP
- D. Anti-dsDNA only

30. A patient has clinical features of celiac disease but negative tissue transglutaminase and antigliadin antibodies. Which genetic test can provide useful supportive information?

- A. HLA-DQ2/DQ8 typing
- B. HLA-B27 only
- C. HLA-A2 typing
- D. ABO genotyping

3.5 Hypersensitivity

31. Seasonal allergen exposure cross-links IgE already bound to mast cells, leading to histamine release. Which hypersensitivity type is this?

- A. Type I
- B. Type II
- C. Type III
- D. Type IV

32. An incompatible ABO transfusion causes antibody-mediated destruction of donor red cells. This is an example of:

- A. Type I
- B. Type II
- C. Type III
- D. Type IV

33. Immune complexes deposit in tissue and activate complement, producing inflammation. Which hypersensitivity category applies?

- A. Type I
- B. Type II
- C. Type III
- D. Type IV

- 34. A tuberculin skin test reaches maximal induration roughly 48-72 hours after exposure. Which cells chiefly mediate this response?**
- A. T lymphocytes and macrophages
 - B. IgE-coated mast cells
 - C. B cells and IgM
 - D. Eosinophils only
- 35. Which test detects immunoglobulin or complement already coating a patient's red cells in vivo?**
- A. Direct antiglobulin test
 - B. Indirect antiglobulin test
 - C. ELISA for total IgE
 - D. Serum protein electrophoresis
- 36. A plasma transfusion triggers severe anaphylaxis in a patient with selective IgA deficiency. Which antibody may be responsible?**
- A. Anti-IgA
 - B. Anti-HBs
 - C. Anti-CCP
 - D. Anti-C3

3.6 Immunoglobulins, Complement, and Cellular Testing

- 37. A young child has recurrent bacterial, fungal, and viral infections. Which initial laboratory study is most appropriate when immune dysfunction is suspected?**
- A. CBC with differential
 - B. Bone marrow biopsy first
 - C. ANA titer
 - D. Tumor marker panel
- 38. A patient repeatedly reacts to plasma-containing products. Which measurement is particularly useful when selective IgA deficiency is suspected?**
- A. Quantitative immunoglobulin levels
 - B. PSA
 - C. Troponin
 - D. RPR
- 39. Which disorder classically produces a monoclonal IgM spike?**
- A. Waldenstrom macroglobulinemia
 - B. DiGeorge syndrome
 - C. C3 deficiency
 - D. Chronic granulomatous disease

40. Which technique is best suited to enumerate mature T-cell subsets in suspected DiGeorge syndrome?

- A. Flow cytometry
- B. RPR
- C. Ouchterlony diffusion
- D. Complement fixation

41. What does a CH50 assay assess?

- A. Functional activity of the classical complement pathway
- B. Total IgE concentration
- C. T-cell proliferation only
- D. Alternative pathway alone

42. Which tests are commonly used to evaluate the phagocyte respiratory burst in chronic granulomatous disease?

- A. DHR or NBT
- B. ANA or anti-dsDNA
- C. SPE or IFE
- D. RPR or TP-PA

3.7 Tumor Testing and Transplantation

43. A colorectal cancer patient has a normal CEA after treatment, followed months later by a sustained increase. What is the most appropriate use of this finding?

- A. Monitor for possible recurrence
- B. Establish the original diagnosis by itself
- C. Determine ABO type
- D. Diagnose autoimmune disease

44. A patient with known ovarian carcinoma has low CA-125 despite documented recurrent disease. What is the best explanation?

- A. A single tumor marker may lack sufficient sensitivity
- B. CA-125 can never be used in ovarian cancer
- C. The recurrence must be benign
- D. The biopsy must be incorrect

45. Why can serum hCG testing be appropriate in a male patient?

- A. Some testicular tumors produce hCG
- B. All adult men normally have pregnancy-level hCG
- C. hCG identifies prostate-specific antigen
- D. It measures HLA compatibility

46. In transplantation, acute rejection occurring about a week after graft placement is primarily associated with which immune effector?

- A. T cells
- B. IgE only
- C. Platelets
- D. Erythrocytes

47. What is a major clinical advantage of reinfusing a patient's own hematopoietic stem cells?

- A. Avoidance of graft-versus-host disease
- B. Creation of stronger alloantibodies
- C. Requirement for more HLA matching
- D. Guaranteed prevention of malignancy recurrence

48. Which molecular approach is widely applicable to HLA typing?

- A. PCR-based testing
- B. Urine dipstick
- C. RPR
- D. Direct antiglobulin testing

3.8 Immunology Problem-Solving

49. Which sequence fails to maintain a constant dilution factor?

- A. 1:4, 1:8, 1:16
- B. 1:5, 1:15, 1:45
- C. 1:2, 1:6, 1:12
- D. 1:3, 1:9, 1:27

50. A patient has a reactive RPR, negative treponemal test, and high anticardiolipin antibody. Which coagulation result may also be abnormal because of lupus anticoagulant activity?

- A. APTT
- B. Bleeding time only
- C. Platelet count necessarily zero
- D. Fibrinogen necessarily absent

51. Within the first few days of acute inflammation, which combination is most expected to increase?

- A. Neutrophils, macrophages, complement, fibrinogen, and C-reactive protein
- B. Only memory B cells and IgG
- C. Only eosinophils and IgE
- D. Plasma cells and anti-ENA antibodies

52. An 18-month-old boy has recurrent severe bacterial infections and suspected Bruton tyrosine kinase deficiency. Which finding is most expected?

- A. Markedly reduced IgG, IgA, and IgM
- B. Markedly increased all immunoglobulins
- C. Isolated loss of CD8 cells only
- D. Normal B-cell maturation with excess plasma cells

53. A patient develops anaphylaxis after several units of plasma and has a history of recurrent respiratory and gastrointestinal infections. Which laboratory investigation best addresses a possible underlying cause?

- A. Quantitative immunoglobulin measurement
- B. PSA measurement
- C. HLA-B27 typing
- D. Cold agglutinin titer only

54. A mixed ANA fluorescence pattern cannot be separated by serial dilution in a patient with overlapping connective-tissue disease features. Which strategy is most useful for defining individual antibody specificities?

- A. Specific antigen assays such as ELISA or multiplex testing
- B. Repeat ABO typing
- C. Measure viral load
- D. Perform a DAT

Answer Key and Rationales

3.1 Basic Principles of Immunology

1. C

Rationale: Immunoglobulins are antigen-specific products of adaptive immunity. Complement and lysozyme participate mainly in nonspecific defense mechanisms.

2. B

Rationale: Hematopoietic cells arise in bone marrow, while T lymphocytes undergo maturation and selection in the thymus.

3. A

Rationale: Activated B cells differentiate into antibody-secreting plasma cells and long-lived memory B cells.

4. A

Rationale: CD3 is characteristically associated with mature T cells and forms part of the T-cell receptor complex.

5. A

Rationale: Cytotoxic CD8 T cells recognize antigen presented by MHC class I molecules, which are expressed on most nucleated cells.

6. C

Rationale: C3 is a central component shared by both pathways and contributes to downstream complement activation.

7. D

Rationale: IgM typically appears first after initial antigen exposure; IgG becomes more prominent in the secondary response.

8. A

Rationale: IgG crosses the placenta through Fc receptor-mediated transport and provides temporary neonatal protection.

3.2 Immunologic Procedures

9. B

Rationale: Affinity describes the strength of a single antigen-antibody interaction; avidity reflects the combined strength of multiple interactions.

10. B

Rationale: In the prozone, antibody excess prevents optimal lattice formation and may produce a false-negative result.

11. B

Rationale: Residual enzyme conjugate can react with substrate even though it is not specifically bound, increasing the measured signal.

12. C

Rationale: A result above the validated calibration range should generally be diluted and reassayed so it falls within the reportable range.

13. A

Rationale: IFE confirms monoclonality and identifies the immunoglobulin heavy- and light-chain classes.

14. B

Rationale: If patient antibody did not fix complement, complement remains available to lyse the indicator red cells; therefore hemolysis represents a negative result.

15. A

Rationale: Gating isolates a population based on scatter or fluorescence characteristics so the intended cells are counted and characterized.

3.3 Infectious Diseases

16. B

Rationale: RPR is a nontreponemal assay and may react in conditions unrelated to syphilis. A negative treponemal test supports a biological false positive.

17. B

Rationale: Fourth-generation assays detect HIV antibodies and p24 antigen, allowing infection to be detected before antibodies are fully developed.

18. A

Rationale: Positive viral RNA with a negative differentiation antibody assay can occur during early infection before seroconversion.

19. B

Rationale: Maternal IgG can persist in the infant and confound antibody testing. Nucleic acid testing directly detects viral material.

20. B

Rationale: HBsAg is an early marker of HBV infection and generally becomes detectable before protective anti-HBs develops.

21. A

Rationale: During the window, HBsAg and anti-HBs may both be absent. IgM anti-HBc remains an important marker of acute infection.

22. A

Rationale: IgM to VCA supports recent infection, while VCA IgG appears during infection and can persist afterward.

23. A

Rationale: Viral load responds relatively quickly to effective or failing therapy and directly reflects the amount of circulating HIV RNA.

3.4 Autoimmune Diseases

24. B

Rationale: Autoimmunity results when immune tolerance to self is lost, allowing immune responses against the body's own antigens.

25. A

Rationale: Anti-Sm is strongly associated with SLE and is considered a highly specific serologic marker.

26. A

Rationale: A speckled ANA pattern commonly reflects antibodies directed against extractable nuclear antigens.

27. A

Rationale: Classic rheumatoid factor is usually IgM that recognizes determinants in the Fc portion of IgG.

28. A

Rationale: Anti-CCP is an important serologic criterion for RA and can be positive even when rheumatoid factor is negative.

29. A

Rationale: Thyroid peroxidase antibodies can occur in autoimmune hyperthyroidism and autoimmune hypothyroidism.

30. A

Rationale: Celiac disease is strongly associated with HLA-DQ2 and/or DQ8. Their presence is supportive, though not diagnostic by itself.

3.5 Hypersensitivity

31. A

Rationale: Type I hypersensitivity is immediate and IgE mediated, with mast-cell degranulation after allergen cross-linking.

32. B

Rationale: Type II hypersensitivity involves antibodies directed against antigens on cells or tissues, resulting in cytotoxic injury.

33. C

Rationale: Type III reactions result from deposition of soluble antigen-antibody complexes followed by complement-driven inflammation.

34. A

Rationale: The tuberculin reaction is a delayed, type IV response driven by sensitized T cells and recruited macrophages.

35. A

Rationale: The DAT detects antibody and/or complement bound to RBCs inside the patient.

36. A

Rationale: IgA-deficient patients can become sensitized to IgA and may develop anti-IgA capable of causing severe reactions after plasma exposure.

3.6 Immunoglobulins, Complement, and Cellular Testing

37. A

Rationale: A CBC and differential provide a practical first screen for abnormal leukocyte numbers or morphology before specialized immune testing.

38. A

Rationale: Quantitative immunoglobulin testing can demonstrate markedly reduced or absent IgA and help explain anti-IgA-associated transfusion reactions.

39. A

Rationale: Waldenstrom macroglobulinemia is a lymphoplasmacytic malignancy characterized by secretion of monoclonal IgM.

40. A

Rationale: Flow cytometry can count and characterize T-cell populations using CD markers, making it useful in thymic/T-cell deficiency.

41. A

Rationale: CH50 measures the serum complement activity required to lyse a defined proportion of antibody-sensitized red cells and reflects classical pathway function.

42. A

Rationale: DHR and the older NBT assay assess oxidative burst activity, which is defective in chronic granulomatous disease.

3.7 Tumor Testing and Transplantation

43. A

Rationale: CEA lacks adequate specificity and sensitivity as a stand-alone diagnostic test, but serial measurements can help monitor treated colorectal cancer for recurrence.

44. A

Rationale: Tumor markers are imperfect. A low CA-125 cannot by itself exclude recurrent ovarian cancer.

45. A

Rationale: Certain testicular tumors secrete hCG, including free beta subunit, so hCG can function as a tumor marker in men.

46. A

Rationale: Acute rejection is predominantly a cell-mediated response in which T lymphocytes attack graft tissue.

47. A

Rationale: Autologous cells originate from the patient, so they do not mount an allogeneic graft-versus-host attack.

48. A

Rationale: HLA genes can be characterized by molecular methods, and PCR-based techniques are commonly used for tissue typing.

3.8 Immunology Problem-Solving

49. C

Rationale: The first step from 1:2 to 1:6 is threefold, but 1:6 to 1:12 is only twofold. A consistent threefold series would continue to 1:18.

50. A

Rationale: Antiphospholipid antibodies may interfere with phospholipid-dependent coagulation assays, commonly prolonging the APTT in vitro.

51. A

Rationale: Early inflammation prominently involves phagocytes, complement, and acute-phase proteins such as fibrinogen and CRP.

52. A

Rationale: Bruton agammaglobulinemia prevents normal B-cell maturation, producing profound deficiency of multiple immunoglobulin classes.

53. A

Rationale: Recurrent mucosal infections plus anaphylaxis to plasma raises concern for IgA deficiency with anti-IgA; measuring immunoglobulins helps evaluate this possibility.

54. A

Rationale: Specific antigen assays can distinguish individual autoantibodies when a complex fluorescence pattern cannot be resolved by dilution.